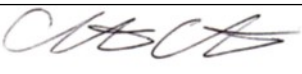


Guideline Title: Intraosseous (IO) Insertion & Infusions

Guideline Number: GUID-3203-PR013

Issue date: 09/09/2014	Replaces Dept. Policy: Intraosseous Infusions (Adult and Pediatric) Using EZ-IO
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To provide an alternative route for administration of intravenous fluids and medications in the adult and pediatric patient when standard peripheral intravenous access has been unsuccessful.
- II. **DEFINITIONS:**
 - a. IO - Intraosseous
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:**
 - i. IV tubing and fluid
 - ii. Proper needle size
 1. 15 mm = pink = 3-39 kg
 2. 25 mm = blue = >39 kg
 3. 45 mm = yellow = >39 kg with excessive tissue over the site
 - iii. Betadine, chlorohexidine, or alcohol prep pads
 - iv. 10 ml aspiration syringe with 5 ml of NaCl
 - v. Lidocaine 2% preservative free injection solution
 - vi. **EZ-IO™ from VidaCare** is approved for patients 3 kg and above as follows:
 1. Proximal tibia: Adults, children, and infants
 2. Distal Tibia: Adults, children, and infants
 3. Proximal humerus: Adults, children, and infants
 4. Distal Femur: Children and infants
 - b. **INDICATIONS:**
 - i. Unsuccessful establishment of peripheral intravenous access or inability to perform venous cannulation as a result of anatomical abnormalities or inadequate positioning.
 - c. **CONTRAINDICATIONS:**
 - i. Ability to obtain IV access by another route (peripheral or external jugular)
 - ii. Fracture or burns to the target bone
 - iii. Previous orthopedic procedures at or proximal to insertion site
 - iv. Infection at the insertion site

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- v. Inability to locate landmarks
- vi. Excessive tissue over the insertion site
- vii. Known congenital defect of the involved bone
- viii. Known bone pathology (cancer or osteoporosis)

d. PROCEDURE:

- i. Always observe body substance isolation procedures and aseptic technique
- ii. Assure that there are no contraindications present.
- iii. Locate the desired insertion site:
 - 1. For the proximal tibia: Locate the patella and the tibial tuberosity. The needle will be inserted 1 cm width medial to the tibial tuberosity. (See Figure 1.)
 - 2. For the proximal humerus: orient the arm at the patient's side, firmly adducted, with the forearm flexed across the umbilicus. Locate the greater tubercle of the humerus. The needle will be inserted laterally, perpendicular to the bone at 90 degrees
- iv. Clean the insertion site with alcohol, chlorohexidine or Betadine. (See Figure 2.)
- v. Prepare the EZ-IO driver and needle set. The needle is sterile. The top of the needle is magnetized and will attach to the end of the driver. Once the needle is attached to the driver, remove the needle from its plastic container.
- vi. Twist the sterile needle cap (cover) and gently remove it from the end of the needle. Do not touch the end of the needle otherwise.
- vii. Place the driver in your dominant hand. Locate your insertion site with the opposite hand.
- viii. Position the needle over the insertion site with the needle at a 90 degree angle to the surface of the bone. Power the needle with firm pressure thru the skin until it reaches the surface of the bone (See Figure 3). There is a 5 mm mark on the needle that should still be visible. If it is not, the procedure should be abandoned as the needle may not be long enough to penetrate the IO space.
- ix. If appropriate, continue the insertion, powering the needle with firm and steady pressure until the flange touches the skin or a sudden lack of resistance is felt (indicating entrance into the bone marrow cavity)
- x. Remove the driver
- xi. While grasping the hub with one hand, rotate the stylet counterclockwise (unscrew it) and remove it. (See Figure 4)
- xii. Attempt to aspirate blood. The needle should be well seated in tibial bone at 90 degrees.
- xiii. Administer Lidocaine 2% preservative free prior to the flush for conscious patients.
 - 1. Adult: **Lidocaine 40 mg IO**
 - 2. Pediatrics: **Lidocaine 0.5mg/kg IO**
- xiv. Flush the needle with 10 ml of sterile saline. There should be a free flow of fluid with no leakage of fluid under the skin.
- xv. Attach the catheter and cover with sterile dressing. (See Figures 5 and 6.)
- xvi. Apply yellow instructional wrist band to patient.

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- xvii. The EZ-IO should not be left in place for more than 24 hours. It can be removed by pulling the hub while rotating gently in a clockwise fashion.

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES:

- a. VidaCare reference materials

Figure 1

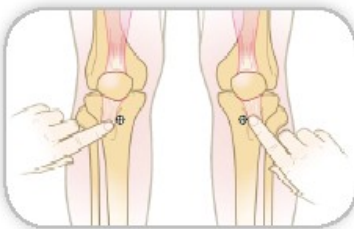


Figure 2



Figure 3



Figure 4

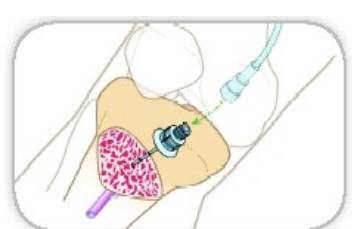


Figure 5

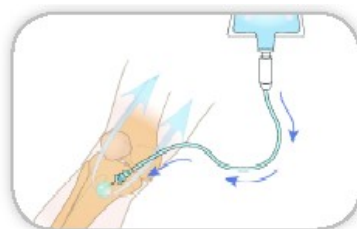


Figure 6

Figure 7

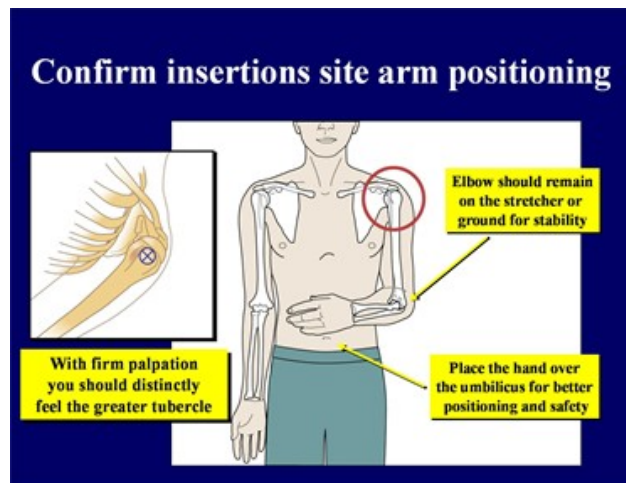


Figure 8

